MIL-F-48306(PA)
19 February 1976
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

FUZE, IMPACT, ATOMIC WEAPON: M1139

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of fuze entitled Fuze, Impact, M1139 which is used in the Lance warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-331	- Fuze and Fuze Components, Environment and Performance Test for.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

ORDNANCE CORPS

8878721	- Spring, Conical.
8879484	- Fuze, Impact, M1139.
8879485	- Receptacle and Plate Assembly.
8879487	- Nut, Internal.

FSC 1135

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the current version before use.

8879489 - Slug.
8879490 - Element, Ceramic - Matched Pair.
8879491 - Wafer, Connecting.
8879492 - Retainer.
8879493 - Insulator.
8879496 - Housing.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The fuze shall comply with all requirements specified on the drawing (dwg) 8879484, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Insulation resistance. The insulation resistance between the inner connector pin and the connector case (with the negative lead to connector case and postive lead to the center pin) shall be 20 megohms minimum at 500 volts direct current (vdc) plus or minus (±) 50 vdc when tested as specified in 4.4.1.

3.6 Capacitance. The capacitance between the inner connector pin and the connector case shall be 110 picofarads (pf) ± 15 pf at a frequency of 1000 Hertz (Hz) ± 50 Hz when tested as specified in 4.4.2.

3.7 Output voltage. The output voltage generated shall be 250 vdc minimum and positive with respect to the impact fuze case and within 160 microseconds (us). Time is defined as the time from initial impact of test fixture to the time the M1139 fuze reaches an output voltage of 250 vdc. During this time period, the fuze shall see an impact of 450 ± 25 gravity units (g's). This test shall be conducted as specified in 4.4.3.

3.8 Environmental testing. The fuze shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The fuze shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the currentversion before use.

3.8.1 Transportation vibration. The fuze shall be subjected to the vibration levels and the longitudinal and transverse axes (2 planes) (see Figure 1) to the vibration levels, at the specified temperatures and for the lengths of times as shown in Table I. The high and low temperatures shall be stabilized (see 6.4) at +165°F, and -65°F, respectively, for 4 hours prior to vibration.

Table I

Frequency (Hz)	Gravity units (g's) or double amplitude (da) -65° + 5°F	+160° + 5°F	Sweep Rate
5 - 15	0.40 ± 0.04 da		3 Frequency sweeps at each temp and each axis. Fifteen minutes each sweep.
15 - 80	5.0 ± 0.5 g		
80 - 115	0.015 ± 0.0015 da	same as -65°F	
115 - 500	10.0 ± 1.0 g		

3.8.2 Operational vibration.

3.8.2.1 Boost phase. The fuze shall be subjected for 10 seconds in the longitudinal and transverse axes (2 planes) (see Figure 1) to a random vibration motion having the following properties: spectral density of 0.1 gravity units squared per Hertz (g2/Hz) between frequencies of 100 Hz to 1000 Hz: 11.00 g's overall root mean square (rms) value. The applied random vibration spectrum shall have a tolerance level of ± 3.0 decibels (Db) between frequencies of 100 and 1000 Hz and a tolerance level of ± 6.0 Db between frequencies of 1000 and 2000 Hz. Attenuation outside the flat portion of the Spectral Density vs Frequency Spectrum shall be 12 Db per octave. The total envelope shall range from 50 to 2000 Hz. This test shall be conducted as specified in 4.4.4.2.1. Upon completion of this test, the fuze shall comply with the requirements of 3.5, 3.6 and 3.7.

3.8.2.2 Sustainer phase. The fuze shall be subjected for 3 minutes in the longitudinal and transverse axes (2 planes) (see Figure 1) to a random vibration motion having the following properties: spectral density 0.02 g2/Hz; 4.92 g's overall rms. The applied vibration spectrum shall have a tolerance level of ± 3.0 Db between frequencies of 1000 and 2000 Hz. Attenuation outside the flat portion of the Spectral Density vs Frequency Spectrum shall be 12 Db/octave. The total envelope shall range from 50 to 2000 Hz. This test shall be conducted as specified in 4.4.4.2.2. Upon completion of this test, the fuze shall comply with the requirements of 3.5, 3.6 and 3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

3.8.3 High temperature. The fuze shall show no evidence of distortion, cracks or swelling, when subjected to a temperature of +160°F ± 5°F as specified in 4.4.4.3. The fuze shall meet the requirements of 3.7 while being at this (+160°F ± 5°F) temperature. Upon satisfactory completion of this test, allow the fuze to return to ambient temperature by cooling in still air at room temperature. Do not use forced cooling by either fan or refrigerated air. The fuze shall then comply with the requirements of 3.5, 3.6 and 3.7 within one (1) hour after having attained ambient room temperature, and tested as specified in 4.4.1, 4.4.2 and 4.4.3.

3.8.4 Low temperature. The fuze shall show no evidence of distortion, cracks or contraction when subjected to a temperature of -65°F ± 5°F as specified in 4.4.4.4. The fuze shall meet the requirements of 3.7 while being at this (-65°F ± 5°F) temperature. Upon satisfactory completion of this test, permit the fuze to return to ambient temperature by allowing it to remain in still air at room temperature. Do not use any forced heating apparatus on the fuze. The fuze shall comply with the requirements of 3.5, 3.6 and 3.7 within one (1) hour after having attained ambient room temperature.

3.8.5 Temperature and humidity. Prior to and following the temperature and humidity test as specified in 4.4.4.5 the fuze assembly shall comply with 3.5, 3.6 and 3.7 within one (1) hour after test and no external deterioration shall exist to the fuze after the test.

3.8.6 Impact shock. The fuze shall be subjected to a shock of 50 gravity units (g's) ± 5g, three times in each direction, along its longitudinal and transverse axis (2 planes) (see Figure 1) for a total of 12 shocks. The duration of each shock shall last for 11 milliseconds (ms) ± 1 ms as measured between points which are 10 percent of the peak level. The time to peak shall be 5.5 ms ± 1 ms. Upon completion of this test, the fuze assembly shall comply with the requirements of 3.5, 3.6 and 3.7, when tested as specified in 4.4.4.6.

3.9 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:55Z
Check the source to verify that this is the currentversion before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 6 complete assemblies (see dwg 8879484) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

Table II

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Insulation resistance (see 3.5)		X
3. Capacitance (see 3.6)		X
4. Output voltage (see 3.7)		X
5. Transportation vibration (see 3.8.1)	X	
6. Operation vibration (see 3.8.2)	X	
7. High temperature (see 3.8.3)		X
8. Low temperature (see 3.8.4)		X
9. Temperature and humidity (see 3.8.5)	X	
10. Impact shock (see 3.8.6)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:55Z
Check the source to verify that this is the current version before use.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer of raw material or components, one production method or manufacturing operation. The entire lot shall be submitted as a unit for inspection.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the current version before use.

Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Fuze, impact, atomic weapon (see dwg 8879484).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper assembly of components	Visual
	102. Inability of connection between wire and wafer to withstand pull test	Gage
	103. Vent hole in receptacle plate soldered prior to soldering receptacle and plate assy (8879485) to housing	Visual
	104. Solder vent hole missing	Visual
	105. Incorrect polarity of ceramic elements (see 3.7)	Visual/Manual
	106. Damaged or loose components	Visual
	107. Improper soldering (see 3.9)	Visual
	108. Evidence of poor workmanship (see 3.10)	Visual
Minor:		
	201. Marking, improper, misleading, illegible or missing (see dwg 8879496)	Visual

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the currentversion before use.

4.3.2.1.2 Housing (see dwg 8879496).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect perpendicularity of bottom surface of smallest inside diameter	Gage
102.	Existence of center nib	Visual/Manual
103.	Incorrect inside diameter for Slug	Gage
104.	Incorrect finish on inside diameter for Slug	Visual
105.	Incorrect concentricity of Slug inside diameter	Gage
106.	Incorrect depth to bottom surface of inside diameter for retainer	Gage
107.	Incorrect depth from bottom surface of retainer inside diameter to bottom surface of smallest inside diameter	Gage
108.	Incorrect hole diameter for straight pin	Gage
109.	Incorrect perpendicularity of straight pin hole diameter	Gage
110.	Incorrect pitch diameter inside tap	Gage
111.	Incorrect pitch diameter outside thread	Gage
112.	Incorrect inside diameter for electrical receptacle plate	Gage
113.	Incorrect perpendicularity of outside shoulder surface (external thread end)	Gage
114.	Incorrect distance across flats	Gage
115.	Incorrect length of hexagon	Gage
116.	Damage of protective finish for solder preform surfaces	Visual
Minor:		
201.	Incorrect depth of finish for Slug inside diameter	Gage
202.	Mislocation of wire sleeve cutout	Gage
203.	Incorrect diameter of wire sleeve cutout	Gage
204.	Incorrect depth of wire sleeve cutout	Gage
205.	Incorrect inside diameter for retainer	Gage
206.	Incorrect depth of largest inside diameter	Gage
207.	Incorrect diameter of shank	Gage
208.	Incorrect total length	Gage

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the currentversion before use.

209. Mislocation of hole for straight pin...Gage
210. Damage to protective finish exposing base metal.....Visual
211. Length of perfect internal threads.....Gage
212. Incorrect clearance diameter for internal nut (8879487).....Gage
213. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.3 Receptacle and plate assembly (8879485).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper assembly of components	Visual
	102. Improper soldering (see 3.9)	Visual
Minor:		
	201. Thread retaining compound missing	Visual
	202. Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.4 Slug (8879489).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect outside diameter	Gage
	102. Incorrect parallelism	Gage
	103. Incorrect perpendicularity	Gage
	104. Improper surface finish (both faces)	Visual
Minor:		
	201. Incorrect length	Gage
	202. Mislocation of vent hole	Gage
	203. Vent hole missing	Visual
	204. Incorrect diameter of vent hole	Gage
	205. Incorrect radius (both ends)	Gage
	206. Evidence of poor workmanship (see 3.10)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T03:55Z
Check the source to verify that this is the current version before use.

4.3.2.1.5 Spring, conical (8878721).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Load required for assembled height incorrect	Gage/Manual
102.	Coils bind when loaded	Visual
103.	Incorrect wind of coils	Visual
104.	Incorrect free length (max)	Gage
105.	Incorrect smaller cone diameter	Gage
106.	Incorrect larger cone diameter (max)	Gage
Minor:		
201.	Ends not ground	Visual
202.	Damage to protective finish exposing base metal	Visual
203.	Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.6 Element, ceramic-matched pair (8879490).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect outside diameter	Gage
102.	Incorrect length	Gage
103.	Incorrect parallelism	Gage
104.	Incorrect perpendicularity	Gage
Minor:		
201.	Incorrect surface finish (both ends)	Visual
202.	Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.7 Wafer, connecting (8879491).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect outside diameter	Gage
102.	Incorrect solder hole diameter (max)	Gage
103.	Mislocation of solder hole diameter	Gage

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the currentversion before use.

104. Incorrect parallelism.....Gage
105. Incorrect protective finish.....Visual
106. Depth of solder hole.....Gage
107. Damage to protective finish exposing base metal.....Visual
108. Incorrect surface finish (both ends).....Visual
109. Evidence of poor workmanship (see 3.10).....Visual

Minor: None defined.

4.3.2.1.8 Retainer (8879492).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Incorrect outside diameter.....Gage
102. Incorrect length of outside diameter...Gage
103. Incorrect depth of smaller inside diameter.....Gage
104. Damage to protective finish exposing base metal.....Visual
105. Incorrect width of slot.....Gage
106. Evidence of poor workmanship (see 3.10).....Visual

Minor: None defined.

4.3.2.1.9 Insulator (8879493).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Incorrect inside diameter.....Gage
102. Incorrect overall length.....Gage
103. Incorrect chamfer (both ends).....Gage
104. Evidence of poor workmanship (see 3.10).....Visual

Minor: None defined.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the currentversion before use.

4.3.2.1.10 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging and packing not in accordance with MIL-STD-1169	Visual
102.	Marking not in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Insulation resistance	Major
b.	Capacitance	Major
c.	Output voltage	Major
d.	Impact shock	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Insulation resistance, capacitance, and output voltage. Each fuze shall be tested in accordance with 4.4.1 thru 4.4.3. If any item fails to comply with 3.5 thru 3.7, the item shall be rejected and removed from the lot.

4.3.3.2 Impact shock. A sample shall be selected at random in accordance with Table V. The sample shall be tested in accordance with 4.4.4.6. If any item fails to comply with 3.8.6, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the currentversion before use.

Table V

Sample sizes for non-destructive testing

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions and calibration instructions designed by the contractor. The contractor shall obtain approval of such designs, operating instructions and calibration instructions prior to use on the contract from Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA- T as specified on DI-R-1714 (see 6.1.2).

4.3.4.2 Minor characteristics and tests. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Insulation resistance. Insulation resistance shall be measured by means of an instrument capable of measuring megohms and current limited to 5 milliamperes (ma). The potential shall be applied for 10 seconds min, and 60 seconds max. The terminals of the impact fuze connector shall be connected as specified in 3.5. Test code shall be 030001.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the current version before use.

4.4.2 Capacitance. The circuit (see 6.3) or its approved equivalent, shall be connected to the impact fuze connector, and the resultant capacitance measured shall be recorded. The capacitance of the test cable shall be subtracted from the capacitance of the cable-impact-fuze combination and the resultant difference shall be recorded as the capacitance of the impact fuze. The impact fuze shall comply with the requirements of 3.6. Test code for the capacitance of the impact fuze shall be 040001".

4.4.3 Output voltage. Test code shall be 050001. The impact fuze shall be mounted on a fixture and the assembly shall be dropped in such a manner so that the impact (small) end of the fuze shall strike normal to a plate surface. A releasing device shall be used and actuated only when the prefall conditions have been met. The output voltage of the impact fuze shall be connected across an impedance load consisting of the following:

A total capacitance of 420 picofarads (pf) ± 20 pf. This capacitance shall include the capacitance of the test cables and the measuring instrument. Each impact fuze assembly shall be dropped 3 times for a distance which, upon impact, shall subject the impact fuze to a deceleration force along its flight axis (x-axis of figure 1) equal to 450 gravity units (g) ± 25 g. The required distance to produce the specified g force (450 g) shall be determined prior to any tests using a calibrated accelerometer. The output voltage for each of the 3 test runs shall be recorded by utilizing an oscilloscope with an impedance of 1 megohm, or higher, and by taking a photograph of the output pulse.

4.4.4 Environmental testing (see 3.8).

4.4.4.1 Transportation vibration.

4.4.4.1.1 Ambient condition. The impact fuze shall be rigidly mounted to a fixture capable of being readily positioned in each of the longitudinal and transverse axes (2 planes). The fixture shall be mounted on a suitable vibration producing apparatus. The impact fuze shall be vibrated at the frequencies, amplitudes, at ambient temperature and for the times specified in Table I. Prior to any vibration test, the impact fuze shall comply with the requirements of 3.5, 3.6, and 3.7. After vibration tests at ambient temperature, the impact fuze shall comply with the requirements of 3.5, 3.6, and 3.7. Dust covers shall be placed on each impact fuze in preparation for vibration testing.

4.4.4.1.2 Low temperature condition (at -65°F). The impact fuze shall be rigidly mounted on a fixture capable of being readily positioned in each of the three mutually perpendicular axes. The fixture shall be mounted on a suitable vibration production apparatus. The impact fuze shall be vibrated at the frequencies, amplitudes, at -65°F, and for the times specified in Table I. Prior to any vibration test, the impact fuze shall comply with the requirements of 3.5, 3.6, and 3.7. After vibration tests at -65°F, the impact fuze shall comply with the requirements of 3.5, 3.6, and 3.7. Dust covers shall be placed on each impact fuze in preparation for vibration testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.

4.4.4.1.3 High temperature condition (at +165°F). The impact fuze shall be rigidly mounted to a fixture capable of being readily positioned in each of the three mutually perpendicular axes. The fixture shall be mounted on a suitable vibration producing apparatus. The impact fuze shall be vibrated at the frequencies, amplitudes, at +165°F and for the times specified in Table I. Prior to any vibration test, the impact fuze shall comply with the requirements of 3.5, 3.6, and 3.7. After vibration tests at +165°F, the impact fuze shall comply with the requirements of 3.5, 3.6, and 3.7. Dust covers shall be placed on each impact fuze in preparation for vibration testing.

4.4.4.2 Operational vibration.

4.4.4.2.1 Boost phase. The impact fuze shall be rigidly mounted to a fixture capable of being readily positioned in the longitudinal and transverse axes (2 planes). The fixture shall be mounted on a suitable vibration producing apparatus and vibrated for the times specified. Following this test, the impact fuze shall be tested as specified in 4.4.1, 4.4.2, and 4.4.3, and comply with the requirements of 3.5, 3.6, and 3.7.

4.4.4.2.2 Sustainer phase. The impact fuze shall be rigidly mounted to a fixture capable of being readily positioned in the longitudinal and transverse axes (2 planes). The fixture shall be mounted on a suitable vibration producing apparatus and vibrated for the times specified. Following this test, the impact fuze shall be tested as specified in 4.4.1, 4.4.2, and 4.4.3, and comply with the requirements of 3.5, 3.6, and 3.7.

4.4.4.3 High temperature. The impact fuze shall be placed in a test chamber with a dust cover on the receptacle. The temperature within the chamber shall be raised to 160°F + 5°F and held stabilized (see 6.4) for 4 hours minimum. The impact fuze shall then be subjected to the test requirement as specified in 3.8.3. Prior to high temperature test, the impact fuze shall comply with the requirements of 3.5 and 3.6.

4.4.4.4 Low temperature. The impact fuze shall be placed in a test chamber with a dust cover on the receptacle. The temperature within the chamber shall be reduced to -65°F + 5°F and held stabilized (see 6.4) for 4 hours minimum. The impact fuze shall then be subjected to the test requirement as specified in 3.8.4. Prior to low temperature test, the impact fuze shall comply with the requirements of 3.5 and 3.6.

4.4.4.5 Temperature and humidity. The impact fuze, with cap, shall be placed within the test chamber and subjected to the temperature and humidity cycle as specified in MIL-STD-331, test 105, for one 14 day cycle in a temperature range of -65°F to +165°F. Upon completion of the last cycle, the impact fuze, removed from the test chamber, moisture removed by wiping, shall comply with requirements specified in 3.8.5.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the currentversion before use.

4.4.4.6 Impact shock. The shock machine utilized shall be capable of producing the specified pulse described in 3.8.6. The shock machine shall be calibrated for conformance with specified wave form specified in above paragraph. Following this test, the impact fuze shall be tested as specified in 4.4.1, 4.4.2, 4.4.3, and comply with 3.5, 3.6, and 3.7.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The fuze shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The fuze shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The fuze shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2026A dated 16 February 1973, and Amendment 1, dated 19 November 1974.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A134

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:55Z
Check the source to verify that this is the currentversion before use.

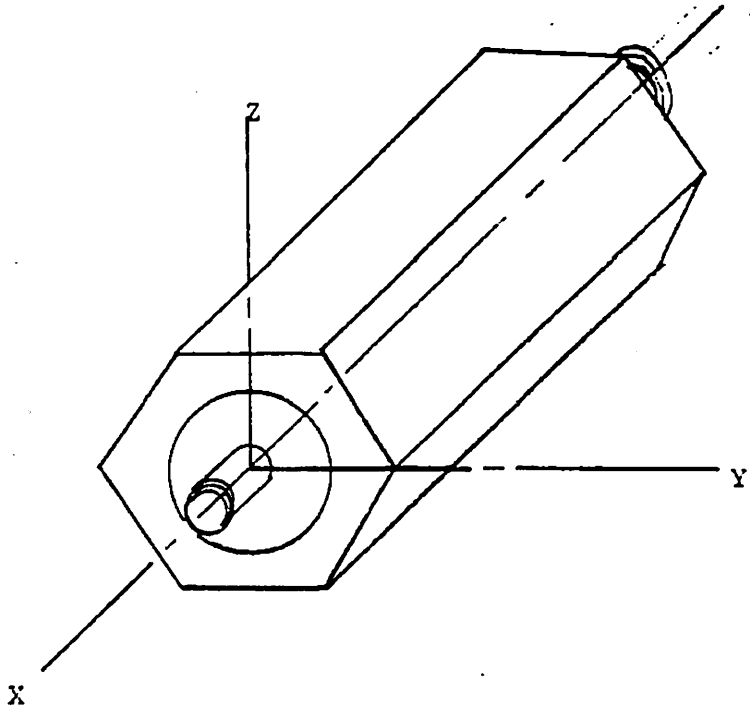


Figure 1 M1139 Impact fuze

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:55Z
Check the source to verify that this is the current version before use.